



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2017/2018

**FOURTH YEAR FIRST SEMESTER EXAMINATION FOR
THE DEGREE OF BACHELOR OF SCIENCE IN PHYSICS
WITH INFORMATION TECHNOLOGY**

MAIN CAMPUS

SPH 401: SOLID STATE PHYSICS

Date: 21st February, 2018

Time: 12.00 - 3.00pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO



SPH 401: SOLID STATE PHYSICS

INSTRUCTIONS

Attempt Question **One** and any other two in this paper.

Useful Constants

Planck's constant $h = 6.62 \times 10^{-34} \text{ Js}$

Electron charge

$$e = 1.6 \times 10^{-19} \text{ C}$$

Electron mass

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

Universal gas constant $R = 8.31 \text{ J / mol / K}$

Boltzman constant

$$K = 1.38 \times 10^{-23} \text{ JK}^{-1}$$

Atomic mass unit

$$1u = 1.66 \times 10^{-27} \text{ kg}$$

QUESTION ONE (30 marks)

- a) i. What is the main difference between the structures of a crystalline and amorphous solid? (2 marks)
- ii. State the conditions required for magnetic resonance to occur in a material (3 marks)
- b) i. Explain briefly how thermal conduction takes place in solids (3 marks)
- ii. Write down the expression for the energy of an elastic wave in a crystal and hence obtain the zero point energy (3 marks)
- c) i. State the limitations of the free electron model (2 marks)
- ii. Mention two causes of electrical resistivity in solid conductors (2 marks)
- d) In each case give an example of a system whose particle properties can be described by the following statistics; Fermi-Dirac, Bose-Einstein and Maxwell-Boltzmann (3 marks)
- e) i. What is Superconductivity? (2 marks)
- ii. State two types of superconductors, explaining clearly the difference between them (2 marks)
- f) i. State Curie's law (2 marks)

ii. Explain why classical models of lattice heat of solids fail to account for the temperature dependence of specific heat in solids at low temperatures (3 marks)

g) Compute the v_{rms} of Oxygen at 0 °C given that the atomic mass of Oxygen is $16u$. (3 marks)

QUESTION TWO (20 marks)

a) i. Write the Schrodinger equation for the electron in a box model (2 marks)

ii. By applying the appropriate boundary conditions to the problem obtain the general solution

(6 marks)

iii. Obtain the expression for the energy eigen values

(6 marks)

b) The total number of electron states from $k=0$ to k in an atom is given by $G(k) = \frac{Vk^3}{3\pi^2}$

where V is the volume of the sample, show that the density of states per unit energy interval

is given by $g(E) = \frac{\frac{3}{2}G(E)}{E}$

(6 marks)

QUESTION THREE (20 marks)

a) Derive the density of states function in the form of

$$\rho(E) = \frac{V}{2\pi} \left(\frac{2m_e}{\hbar^2} \right)^{3/2} E^{1/2} \quad (6 \text{ marks})$$

b) Assuming that the energy of the atoms in a crystal is governed by the Maxwell- Boltzmann statistics.

Show that the mean energy at high temperature is given by

$$E \approx kT \quad (14 \text{ marks})$$

QUESTION FOUR (20 marks)

a) A gold crystal has a face centered cubic lattice with one atom per lattice point and a lattice constant

$4.08 \text{ \AA}$, every gold atom having one valence electron. Estimate

i. the electron Fermi energy

(8 marks)

ii. Fermi velocity

(4 marks)

b) The energy of N oscillators in the Einstein's model of a solid is given by $E = N\langle n \rangle \hbar \omega$

where $\langle n \rangle$ is the Bose-Einstein distribution function. Show that the molar specific capacity at

constant volume is given by
$$C_v = 3Nk(\beta \hbar \omega)^2 \frac{e^{\beta \hbar \omega}}{(e^{\beta \hbar \omega} - 1)^2}$$

(Assume each of the n atoms has $3N$ degrees of freedom)

(8 marks)

QUESTION FIVE (20 marks)

a) Briefly describe the following magnetic properties

- Diamagnetism
- Ferromagnetism
- Paramagnetism

(8 marks)

b) i. Show that the magnetic field inside a solenoid with a soft iron core is given by $B = \mu_o \mu_r H$

where μ_o is permeability of free space, μ_r is relative permeability

(6 marks)

ii. Derive an expression for the Hall constant R_H

(3 marks)

iii. Determine the value of Hall constant for a sample of P-type germanium of conductivity $100 \Omega^{-1} m^{-1}$

(take $\mu_e = 0.39 m^2 V^{-1} s^{-1}$ and $\mu_h = 0.19 m^2 V^{-1} s^{-1}$)

(3 marks)



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**FOURTH YEAR FIRST SEMESTER EXAMINATION FOR
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MAIN CAMPUS

SPH 402: STATISTICAL MECHANICS

Date: 19th February, 2018

Time: 12.00 - 3.00pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO in Section B.



Answer question I (Compulsory) and any other TWO questions in section B.

SECTION A

QUESTION 1 (30 MARKS)

- (a) Define the following terms as used in Statistical Mechanics
- (i) Phase space (1 mark)
 - (ii) Density of states (1 mark)
 - (iii) Microstate (1 mark)
 - (iv) Statistical weight (1 mark)
- (b) State any TWO properties of particles obeying Fermi-Dirac distribution (2 marks)
- (c) Determine the statistical weight of 4 bosons in an energy level of degeneracy 3. (3 marks)
- (d) Define the term degree of freedom, hence state the number of degrees of freedom of an assembly of N particles in a classical phase space. (2 marks)
- (e) The Helmholtz free energy of a gas is given by the equation
- $$F = N\varepsilon_0 - Nk_B T \ln\left(\frac{eV}{N}\right) - NcT \ln T - NbT$$
- Here $e = 2.718...$ is the base of the natural logarithms, N is the number of particles, V is the volume, T is the temperature while ε_0 , b and c are constants. Calculate:
- (i) The entropy as a function of V and T . (3 marks)
 - (ii) The internal energy U as a function of T and N . (3 marks)
- (f) Starting with the definition of Gibb's free energy, derive expression for chemical potential. (3 marks)
- (g) Derive the equation for the partition function of a spin- $\frac{1}{2}$ paramagnetic system. (3 marks)

- (h) Explain the term black body as used in Statistical Physics. (2 marks)
- (i) Explain the term Bose-Einstein condensation. (2 marks)
- (j) Show that the conditions for equilibrium of a system in a heat bath can be derived from the inequality $dU_A - T_R dS_A + P_R dV_A - \mu_R dN_A \leq 0$ where the symbols have their usual meanings. (3 marks)

SECTION B

Answer any TWO questions in this section

QUESTION 2 (20 MARKS)

- (a) Derive Hamilton's equation for a dynamical system in a phase space. (10 marks)
- (b) (i) Show that when all states of a system are equally likely (all the probabilities, p_i are equal) the maximum entropy is obtained where $S = -\sum_{i=1}^n p_i \ln p_i$ and $\sum_{i=1}^n p_i = 1$ (6 marks)
- (ii) State the physical significance of the result in 2 b(i). (1 mark)
- (c) By defining $x = \left(\frac{\epsilon_i - \mu}{k_B T} \right)$ in the Fermi-Dirac distribution function, show that the average population of an eigenstate for fermions cannot exceed 1. (3 marks)

QUESTION 3 (20 MARKS)

- (a) Show that the Bose-Einstein distribution function is given by $f_{BE} = \left(e^{\left(\frac{\epsilon_i - \mu}{k_B T} \right)} - 1 \right)^{-1}$ (10 marks)
- (b) Show that the Stirling's approximation for a very large number of particles, N , is given by $\ln N! = N \ln N - N$. (5 marks)
- (c) Calculate the percentage error in the use of Stirling's approximation to calculate $\ln N!$ for a system of $N = 10000$ particles. (5 marks)

QUESTION 4 (20 MARKS)

- (a) Determine the conditions under which a composite isolated system C resulting from interaction of two isolated systems A and B attains thermodynamic equilibrium. **(13 marks)**
- (b) Obtain the equation for the internal energy of a system consisting of N independent particles in terms of Boltzmann constant, k_B , temperature, T and partition function, Z . **(7 marks)**

QUESTION 5 (20 MARKS)

- (a) The surface temperature of the Sun is 5700 K and the spectrum of radiation which it emits has a maximum at a wavelength of 510 nm . Estimate the surface temperature of the North Star, for which the corresponding maximum is at 350 nm . **(5 marks)**
- (b) Black body radiation in a box of volume V and at temperature T has internal energy $U = \sigma VT^4$ and pressure $P = \frac{1}{3}\sigma T^4$ where σ is the Stefan-Boltzmann constant. Determine the Helmholtz free energy, entropy, pressure and chemical potential in terms of U . **(12 marks)**
- (c) Outline any THREE characteristics of Maxwell-Boltzmann statistics. **(3 marks)**



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**FOURTH YEAR FIRST SEMESTER EXAMINATION FOR
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MAIN CAMPUS

SPH 405: PHYSICS OF MATERIALS

Date: 15th February, 2018

Time: 3.30 - 6.30pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO



SECTION A: Question 1 of this section is compulsory.

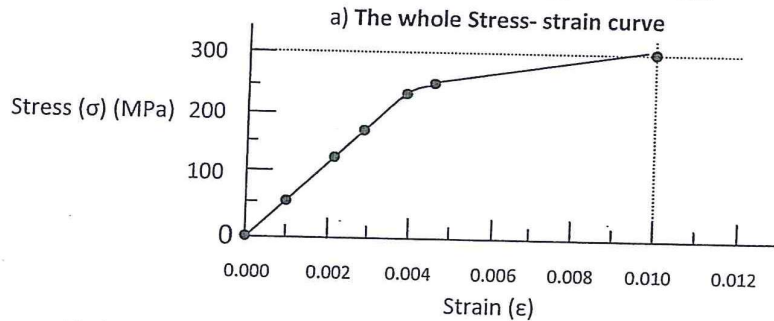
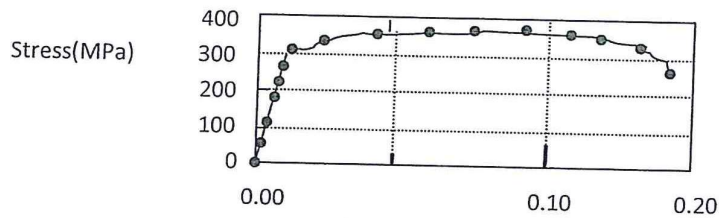
It carries **30 marks**.

1. a) (i) Give an equation for critical fibre length (l_c) for effective stiffening and strengthening in a composite material and define the parameters in the equation. **(4 marks)**
(ii) Explain why longer, thinner fibres are preferred to shorter, thicker fibres in composites. **(3 marks)**
- b) (i) Explain the following constants that are associated with the fracture in a material: k_t , K_t and K_{IC} **(6 marks)**
(ii) Why is *fracture surface energy* greater than *intrinsic surface energy*? **(1 mark)**
- c. (i) Define viscosity. **(1 mark)**
(ii) A glass slab at room temperature, with dimensions 140mm x 60mm x 20mm, has a shear force (F_t) of 8.4 kN across its largest faces. The viscosity of this glass at room temperature is 10^{16} Pa s and the average separation between molecules is 0.5nm. Find the time for two neighbouring molecules to slide past each other. **[4 marks]**
- d) (i) What is *rule of mixtures* as applied to composites? **(2 marks)**
(ii) Cite three properties of a composite material that the above rule can predict. **(3 marks)**
- e) (i) Light absorption in materials is measured in terms of *absorption coefficient* (α_a). Give the equation for α_a and define the unknowns. **(3 marks)**
(ii) Diamond has refractive index of 2.41. Compute the percentage of light that can be reflected from its surface. **(3 marks)**

SECTION B: Answer **ANY TWO** questions from this section.
Each question carries **20 marks**.

2. a) Define *ductility* and *resilience*. (4 marks)

b) A cylindrical specimen of aluminium alloy having a diameter of 12.800 mm and a gauge length of 50.800 mm is pulled in tension. The *stress verses strain* graphs from the loading are given below.



b) The same Stress-strain curve extending just beyond the elastic region deformation

- Compute the modulus of elasticity (E). (4 marks)
- Determine the yield strength at a strain offset of 0.002. (2marks)
- Determine the tensile strength of this alloy. (2 marks)
- Calculate the approximate ductility, in percentage elongation. (4 marks)
- Compute the modulus of resilience. (4 marks)

- Explain and distinguish between the terms *strength* and *toughness*. (4 marks)
 - Describe ways of measuring both of these quantities for metallic alloys. (4 marks)

b) The stress intensity K_1 in mode 1 loading in an infinite plate containing a centre-crack of length $2a$ is given by

$$K_1 = \sigma_{app}(\pi a)^{1/2} \dots\dots\dots(1)$$

where σ_{app} is the applied stress.

The corresponding relationship for a semi-infinite plate containing an edge crack of length a is

$$K_1 = 1.12\sigma_{app}(\pi a)^{1/2} \dots\dots\dots(2)$$

Comment on why it is necessary to include a factor 1.12 in the second equation. (2 marks)

- Define fracture? (2 marks)
 - When does fracture take place in a material? (2 marks)

- (iii) I) A very large plate of a high strength aluminium alloy contains an edge crack of length 40 mm. Given that it fails at an applied stress of 100 MPa, calculate the fracture toughness of the alloy. (3 marks)
- II) How would the above calculation be modified if the edge crack occurred at a large circular hole machined in the plate? (3 marks)

- 4) a) (i) Heat travels at different rates in different materials. Give the equation for rate of heat transfer (H). Define the parameters used. (4 marks)
- (ii) When is something neither hot nor cold? (1 mark)
- (ii) An aluminium pot contains water that is kept steadily boiling (100°C). The bottom surface of the pot, which is 12 mm thick and $1.5 \times 10^4 \text{ mm}^2$ in area, is maintained at a temperature 102°C by an electric heating unit. Find the rate at which heat is transferred through the bottom surface. [Thermal conductivity for aluminium (k_{AL}) = $235 \text{ Wm}^{-1}\text{K}^{-1}$] (3 marks)

- b) (i) What is the basic equation for *thermal conductivity* (k) in terms of *specific heat*? (1 mark)
- (ii) Cite two practical importance of k. (2 marks)
- (iii) What role do phonons play in k? (1 mark)
- (iv) What effect do specimen size, grain structure and phase structure have on k? (2 marks)

- c) How many 20 gm ice cubes, whose initial temperature is -10°C, must be added to 1 litre of hot tea, whose initial temperature is 90°C, in order that the final temperature of the mixture is 10°C? (6 marks)

Note: Latent heat of fusion of ice (L_v) = 333 kJ. Kg^{-1}

Specific heat of water (c_w) = $4190 \text{ J.kg}^{-1}\text{K}^{-1}$

Specific heat of ice (c_i) = $2100 \text{ J.kg}^{-1}\text{K}^{-1}$

Assume that the tea has the same properties as water; 1 litre of water has a mass of 1 kg.

5. a) (i) Define a *composite*. (2 marks)
- (ii) Composite stiffness can be predicted using micro-mechanics approach termed the rule of mixtures. State three assumptions you must apply before using this rule. (3 marks)
- (iii) In a composite, one material is continuous phase and the other discontinuous phase. What names are given to them and what are their relative strengths? (3 marks)
- (iv) Cite three functions of the matrix phase in a composite. (3 marks)

- b) Consider a composite material being loaded parallel to the fibres and under uniaxial tension σ_∞ . Derive the overall elastic modulus of the composite E_c . (6 marks)

- c) Give three disadvantages of composites. (3 marks)

END



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FOURTH YEAR FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN PHYSICS WITH INFORMATION TECHNOLOGY

MAIN CAMPUS

SPH 406: DIGITAL ELECTRONICS II

Date: 20th February, 2018

Time: 12.00 - 3.00pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO in Section B.



INSTRUCTIONS: Question 1 of section A is **COMPULSORY**.

Attempt **ANY TWO** questions from section B.

SECTION A: This section is **COMPULSORY**.

1. a) You need to reset (with a HIGH signal) your microprocessor when power is first applied to it. Given resistors ($R_1 = 1\text{M}\Omega$ and $R_2 = 1\text{ k}\Omega$), capacitor ($1\mu\text{F}$) and inverting buffer (7474);

(i) wire a circuit that will do the job. **(3 marks)**

(ii) name and define the circuit. **(3 marks)**

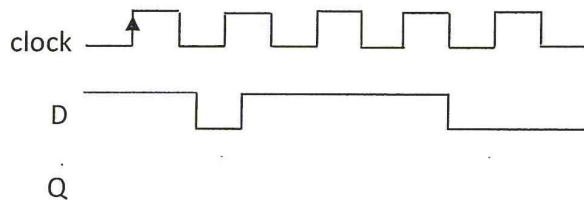
b) (i) State commonly used display techniques. **(2 marks)**

(ii) Give four important criteria for choosing a display technique. **(4 marks)**

c) (i) Define a *flip-flop*. **[2 mark]**

(ii) Name three types of wiring errors that could cause a flip-flop to operate improperly. **[3 marks]**

(iii) The signal D in figure 1 is input to a D-type flip-flop as shown in figure 1. Draw the corresponding Q output. Q is initially LOW. **[3 marks]**



d) (i) What is a *pulse width* of a monostable? **(2 marks)**

(ii) Name the components that are used to control the pulse width. **(2 marks)**

(iii) What parameter determines the duration of the pulse width? **(1 mark)**

e) (i) Explain what is meant by the *delay mode* operation of a 555 timer. **(1 mark)**

(ii) If $V_{CC} = 5\text{V}$, find the threshold (V_{Thres}) and trigger (V_{Trig}) voltages of the timer. **[4 marks]**

SECTION B: Answer **ANY TWO** questions from this section. Each question carries 20 marks.

2. a) (i) What is a *register word*? [1 mark]

(ii) Differentiate between a *parallel-in-serial-out* and *serial-in-parallel-out* register. [4 marks]

b) Figure 2 shows the contents of a register.

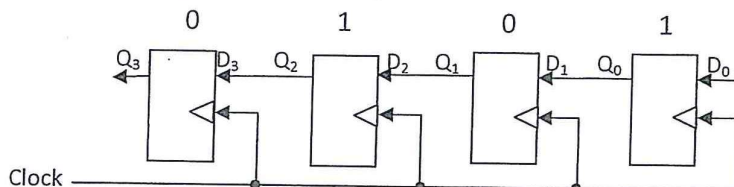


Fig. 2

Give the contents of the register after:

(i) the first clock pulse

(ii) the second clock pulse

(iii) the third clock pulse [3 marks]

c) (i) Describe the operation of a binary counter. [3 marks]

(ii) State main sections of a frequency counter and name a main factor that determines its accuracy. [3 marks]

d) Using a 4-bit binary counter, draw a block diagram of a circuit that will give a period of an unknown waveform. Explain its operation. [6 marks]

3. a) Explain why it is good engineering practice to construct digital systems so that they operate synchronously. Comment on the clock requirement of such a system. (6 marks)

b) (i) Figure 3 shows waveforms of a 4-bit synchronous counter. Assuming that the fourth waveform is to be produced from JKFF, what conditions have to be satisfied for the output of the flip-flop to toggle. (3 marks)

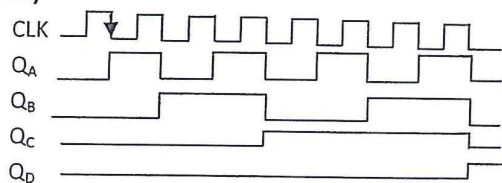


Fig. 3

(ii) Draw the circuit that implements the 4-bit counter above. Use JK flip-flops. (5 marks)

c) A logic circuit is shown in figure 4.

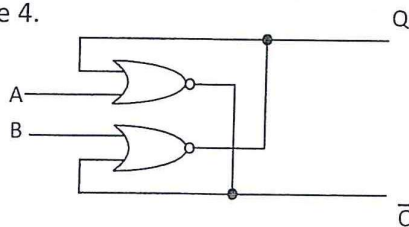


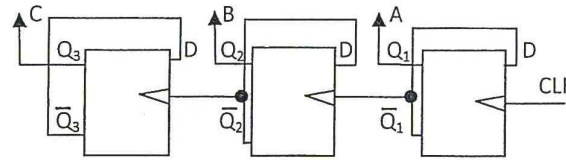
Fig.4

(i) What is the name of the circuit? (1 mark)

(ii) Assuming that initially $Q = 0$ and that A and B are both zero, explain what happens if both inputs go to HIGH (i.e one). (5 marks)

4. a) Using a properly labelled diagram, give an equivalent circuit using switches that describe the operation of an ordinary flip-flop. (4 marks)

b) The diagram in figure 8 shows a counter.



- (i) Name it. (1 mark)
- (ii) Why is named so? (2 marks)
- (iii) Explain its operation. (7 marks)

- c) (i) Give a block diagram of a master-slave JK flip-flop. (5 marks)
- (ii) What item in the circuit makes this flip-flop different from the ordinary JK flip-flop? (1 mark)

5. a) (i) You are given three similar D-type flip-flops. Use them to design asynchronous 3-bit binary counter wired in a toggling mode. [4 marks]
- (ii) Complete Table 1 for the outputs Q_0 (LSB), Q_1 and Q_2 (MSB) from the counter for the clock pulses shown. Assume the initial state of the counter is zero. [3 marks]

Clock	Q_0	Q_1	Q_2
1 st pulse			
2 nd pulse			
3 rd pulse			
4 th pulse			
5 th pulse			
6 th pulse			
7 th pulse			

Table 1

- (iii) Give the reading of the counter at the 9th and 10th pulses? [2 marks]
 - (iv) Deduce an expression for the output frequency f_n for any flip-flop in the counter. Define your unknowns. [3 marks]
- b) (i) Define a *tri-state logic*. [2 marks]
- (ii) Explain what happens when a device switches to a *z-state* and why this is important. [4 marks]
- c) Define a *circuit buffer*. (2 marks)

END